Cells and Their Organelles

It's just like…a prison cell!

Definition of 'cell'

- The smallest basic unit of a plant or animal. (Organism)
- A small room that does not have much furniture. E.g. a prison or a monastery or convent. (Room)
- A small component making part of a whole: *the cells of a honeycomb.* (Part)

The cells we study in Biology are the smallest basic unit that makes up plants and animals. Similarly, a 'cell' is also the name of a small, contained space that, when joined with others, makes up a larger structure. However, are there similarities between these different types of cells, in terms of the parts they contain? Let's take a look.

Below is a list of some of the main parts of a cell found in an organism. Compare these to the parts of another cell type of your choice. (E.g. a prison, monastery or convent cell, honeycomb cell or any other type of cell). You could identify an analogy for each cell part or choose one cell part and write a persuasive text describing how it has a similar meaning and/or function in different types of cells.

Cells (organism)		**Cells (other type e.g. prison)**	
Cell part	**Description and function in a cell**	**Is like a _____** *e.g. prison*	**Details**
cell wall	Provides rigidity; protects cell contents; allows *some* materials to pass in and out.	*Prison wall*	*Keeps things inside; controls what/who enters, e.g. prison guards, food. Stops other things from escaping, e.g. prisoners.*
cytoplasm	Everything outside of the nucleus.		
nucleus	The cell's brain or control room. Where the DNA is found. The activities of the cells are directed from here.		

17SCI01A© ClickView Pty Limited 2017.

Cells and Their Organelles

It's just like…a prison cell!

Cells (organism)		Cells (other type e.g. prison)	
Cell part	**Description & function in a cell**	**Is like a _____** *e.g. prison*	**Details**
endoplasmic reticulum	Tubes through which materials are transported throughout the cell. The cellular highway.		
ribosomes	The protein-making machines.		
Golgi complex	Processes and synthesises the proteins, which are then moved to other parts of the cell for strengthening and building.		
mitochondria	Convert sugar to ATP, the most useful souce of energy for the cell.		
lysosomes	Engulfs substances not needed by the cell and recycles cellular waste.		
vacuole	Storage place for water, nutrients and wastes.		

17SCI01A© ClickView Pty Limited 2017.